

5th Cryptography Homework

6.8.12 You are trying to factor $n = 642401$. Suppose you discover that

$$516107^2 = 7 \pmod{n}$$

$$187722^2 = 2^2 \cdot 7 \pmod{n}$$

Use this information to factor n .

$$(516107 \cdot 187722)^2 = 2^2 \cdot 7^2 \pmod{612401}.$$

$$(289038)^2 = (14)^2 \pmod{612401}.$$

$$\gcd(289038 - 14, 642401) = 1129.$$

The other factor is $642401/1129 = 569$.

6.8.23 Your opponent use RSA with $n = pq$ and encryption exponent e and encrypts a message m . This yields the ciphertext $c = m^e \pmod{n}$. A spy tells you that, for this message, $m^{12345} = 1 \pmod{n}$. Describe how to determine m . Note that you do not know $p, q, \phi(n)$, or the secret decryption exponent d . However, you should find a decryption exponent that works for this particular ciphertext. Moreover, explain carefully why your decryption works.

We know that $m^{12345} = 1 \pmod{n}$ and we find d s.t. $de = 1 \pmod{12345}$. The message $m = c^d = m^{ed \pmod{12345}} = m^1 \pmod{n}$.

6.8.25 You may assume the fact that $m^{270300} = 1 \pmod{1113121}$ for all m with $\gcd(m, 1113121) = 1$. Let e and d satisfy $ed = 1 \pmod{270300}$, and suppose that m is a message such that $0 < m < 1113121$ and $\gcd(m, 1113121) = 1$. Encrypt m as $c = m^e \pmod{1113121}$. Show that $m = c^d \pmod{1113121}$. Show explicitly how you use the fact that $ed = 1 \pmod{270300}$ and the fact that $m^{270300} = 1 \pmod{1113121}$. (Note: $\phi(1113121) \neq 270300$, so Euler's theorem does not apply.)

$$m = c^d = m^{ed} = m^{270300} \cdot m^{270300} \cdots m^{270300} \cdot m^{(ed \pmod{270300})}$$

$$= 1 \cdot 1 \cdots 1 \cdot m^1 = m \pmod{n}.$$

6.8.27.a Suppose Alice wants to send a short message m but wants to prevent the short message attack of Section 6.2. She tells Bob that she is adjoining 100 zeros at the end of her plaintext, so she is using $m_1 = 10^{100}m$ as the plaintext and sending $c_1 = m_1^e$. If Eve knows that Alice is doing this, how can Eve modify the short plaintext attack and possibly find the plaintext?

$$c_1 = m_1^e = (10^{100}m)^e \pmod{n}.$$

Eve makes two list:

$$1. c_1 x^{-e} \text{ for all } x \text{ with } 1 \leq x \leq 10^9$$

$$2. 10^{100e} y^e \text{ for all } y \text{ with } 1 \leq y \leq 10^9$$

If she finds one match, then she has $c_1 x^{-e} = 10^{100e} y^e$ for some x, y .
This yields

$$c_1 = x^e 10^{100e} y^e = 10^{100e} x^e y^e = (10^{100} xy)^e \pmod{n},$$

so $m = xy \pmod{n}$.

6.8.27.b Suppose Alice realizes that the method of part (a) does not provide security, so instead she makes the plaintext longer by repeating it two times: $m||m$ (where $x||y$ means we write the digits of x followed by the digits of y to obtain a longer number). If Eve knows that Alice is doing this, how can Eve modify the short plaintext attack and possibly find the plaintext? Assume that Eve knows the length of m . (Hint: Express $m||m$ as a multiple of m .)

We denote the length of m as $|m|$.

$$c = (m||m)^e = (m \times 10^{|m|} + m)^e = 10^{|m|e} m^e + m^e = (10^{|m|e} + 1) m^e \pmod{n}.$$

Eve makes two list:

$$1. c x^{-e} \text{ for all } x \text{ with } 1 \leq x \leq 10^9$$

$$2. (10^{|m|e} + 1) y^e \text{ for all } y \text{ with } 1 \leq y \leq 10^9$$

If she finds one match, then she has $c x^{-e} = (10^{|m|e} + 1) y^e$ for some x, y . This yields

$$c = x^e (10^{|m|e} + 1) y^e = (10^{|m|e} + 1) x^e y^e = 10^{|m|e} (xy)^e + (xy)^e \pmod{n},$$

so $m = xy \pmod{n}$.

7.6.2.a Compute $6^5 \pmod{11}$

$$6^5 = 6^2 \pmod{11} \cdot 6^2 \pmod{11} \cdot 6 \pmod{11} = 3 \cdot 3 \cdot 6 \pmod{11} = 10 \pmod{11}.$$

7.6.2.b Let $p = 11$. Then 2 is a primitive root. Suppose $2^x = 6 \pmod{11}$. Without finding the value of x , determine whether x is even or odd.

2 is a primitive root. $2^{10} = 1 \pmod{11}$.

$$2^5 = -1 \pmod{11} \text{ and } 6 = 2^x \pmod{11}.$$

$$6^5 = (2^x)^5 = 10 = -1 \pmod{11}.$$

Therefore, x is odd.

7.6.3 It can be shown that 5 is primitive root for the prime 1223. You want to solve the discrete logarithm problem $5^x = 3 \pmod{1223}$. Given that $3^{611} = 1 \pmod{1223}$, determine whether x is even or odd.

Since 5 is a primitive root for the prime 1223

$$5^{1222} = 1 \pmod{1223}.$$

$$5^{611} = -1 \pmod{1223}.$$

$$3^{611} = (5^x)^{611} = 1 \pmod{1223}.$$

Therefore, x is even.

7.6.7 Suppose you know that

$$3^6 = 44 \pmod{137}, 3^{10} = 2 \pmod{137}$$

Find a value of x with $0 \leq x \leq 135$ such that $3^x = 11 \pmod{137}$

We have $3^{10} = 2 \pmod{137}$, raise both sides to -2 power to obtain $(3^{10})^{-2} = 2^{-2} = 4^{-1} \pmod{137}$. We have

$$3^6 \cdot 3^{-20} = 3^{-14} = 44 \cdot 4^{-1} = 11 \pmod{137}.$$

Hence, $3^{-14} = 3^{122} = 11 \pmod{137}$, $x = 122$.

7.6.9 Let's reconsider Exercise 22 in Chapter 3 from the point of view of Pohlig-Hellman algorithm. The only prime q is 2. For k as in that exercise, write $k = x_0 + 2x_1 + \cdots + 2^{15}x_{15}$.

7.6.9.a Show that the Pohlig-Hellman algorithm yields

$$x_0 = x_1 = \cdots = x_{10} = 0$$

and

$$2 = \beta = \beta_1 = \cdots = \beta_{11}$$

See 7.6.9.b.

7.6.9.b Use the Pohlig-Hellman algorithm to compute k .

$\alpha = 3, \beta = 2, p = 65537$. Note that $p - 1 = 65537 - 1 = 2^{16}$. Write $k = x_0 + 2x_1 + \cdots + 2^{15}x_{15}$.

$$\beta^{(p-1)/2} = 2^{32768} = 1 \pmod{65537}, \alpha^{(p-1)/2} = 3^{32768} = -1 \pmod{65537}.$$

Since $\beta^{(p-1)/2} = (\alpha^{(p-1)/2})^{x_0}$, we have $x_0 = 0$. Next,

$$\beta_1 = \beta \alpha^{-x_0} = 2 \pmod{65537}, \beta_1^{(p-1)/2^2} = 1 \pmod{65537}, x_1 = 0.$$

$$\beta_2 = \beta_1 \alpha^{-2x_1} = 2 \pmod{65537}, \beta_2^{(p-1)/2^3} = 1 \pmod{65537}, x_2 = 0.$$

$$\beta_3 = \beta_2 \alpha^{-2^2 x_2} = 2 \pmod{65537}, \beta_3^{(p-1)/2^4} = 1 \pmod{65537}, x_3 = 0.$$

$$\beta_4 = \beta_3 \alpha^{-2^3 x_3} = 2 \pmod{65537}, \beta_4^{(p-1)/2^5} = 1 \pmod{65537}, x_4 = 0.$$

$$\begin{aligned}
\beta_5 &= \beta_4 \alpha^{-2^4 x_4} = 2 \pmod{65537}, \beta_5^{(p-1)/2^6} = 1 \pmod{65537}, x_5 = 0. \\
\beta_6 &= \beta_5 \alpha^{-2^5 x_5} = 2 \pmod{65537}, \beta_6^{(p-1)/2^7} = 1 \pmod{65537}, x_6 = 0. \\
\beta_7 &= \beta_6 \alpha^{-2^6 x_6} = 2 \pmod{65537}, \beta_7^{(p-1)/2^8} = 1 \pmod{65537}, x_7 = 0. \\
\beta_8 &= \beta_7 \alpha^{-2^7 x_7} = 2 \pmod{65537}, \beta_8^{(p-1)/2^9} = 1 \pmod{65537}, x_8 = 0. \\
\beta_9 &= \beta_8 \alpha^{-2^8 x_8} = 2 \pmod{65537}, \beta_9^{(p-1)/2^{10}} = 1 \pmod{65537}, x_9 = 0. \\
\beta_{10} &= \beta_9 \alpha^{-2^9 x_9} = 2 \pmod{65537}, \beta_{10}^{(p-1)/2^{11}} = 1 \pmod{65537}, x_{10} = 0. \\
\beta_{11} &= \beta_{10} \alpha^{-2^{10} x_{10}} = 2 \pmod{65537}, \beta_{11}^{(p-1)/2^{12}} = -1 \pmod{65537}, x_{11} = 1. \\
\beta_{12} &= \beta_{11} \alpha^{-2^{11} x_{11}} = 16384 \pmod{65537}, \beta_{12}^{(p-1)/2^{13}} = -1 \pmod{65537}, x_{12} = 1. \\
\beta_{13} &= \beta_{12} \alpha^{-2^{12} x_{12}} = 256 \pmod{65537}, \beta_{13}^{(p-1)/2^{14}} = 1 \pmod{65537}, x_{13} = 0. \\
\beta_{14} &= \beta_{13} \alpha^{-2^{13} x_{13}} = 256 \pmod{65537}, \beta_{14}^{(p-1)/2^{15}} = -1 \pmod{65537}, x_{14} = 1. \\
\beta_{15} &= \beta_{14} \alpha^{-2^{14} x_{14}} = 65536 \pmod{65537}, \beta_{15}^{(p-1)/2^{16}} = -1 \pmod{65537}, x_{15} = 1.
\end{aligned}$$

Hence, $k = 55296$.